TWO NONZERODIVISORS IN A ONE-DIMENSIONAL LOCAL RING ACQUIRE A COMMON FACTOR AFTER A FINITE BLOWUP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros) of sources/stacks-chow/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. All five referenced results live in earlier chapters, so their locators point at the verbatim lines of the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt) that carry the canonical tags.

Editorial note. This is the first card using the new lane ident range L800-L999 (Chief 2026-10-12), which ends the cross-lane numbering collisions.

Allowed standard prerequisites (editorial).

- Blowing up in an ideal, and the effective Cartier divisor it produces. Divisors, Lemma labelled divisors-lemma-blowing-up-gives-effective-Cartier-divisor (canonical Stacks tag 02OS); invoked at proof lines 581-582 and again at 596. Supplies that the blowup X = Proj(oplus I^d) -> Spec(R) is an isomorphism over the punctured spectrum and that IR' is locally principal.
- The blowup is projective. Divisors, Lemma labelled divisors-lemma-blowing-up-projective (canonical Stacks tag 02NS); invoked at proof lines 586-587. Supplies projectivity, which is combined with the next two inputs to conclude finiteness.
- Generic points of the blowup and finiteness of a morphism finite in codimension one. Divisors, Lemma divisors-lemma-blow-up-and-irreducible-components (0BFM, line 589) and Varieties, Lemma varieties-lemma-finite-in-codim-1 (0AB7, line 590). Supply that every generic point of X lies over the punctured spectrum, so that the projective morphism is finite.
- A locally free module of rank one over a semi-local ring is free. Algebra, Lemma labelled algebra-lemma-locally-free-semi-local-free (canonical Stacks tag 02M9); invoked at proof lines 598-599. Supplies that the locally principal ideal IR' is in fact principal, which produces the common factor t.

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Source reference. Original lemma 02OS: sources/stacks-chow/tags.txt, source lines 3527–3527. Cross-chapter standard prerequisite invoked twice, at proof lines 582 and 596, for the behaviour of the blowup over the punctured spectrum and local principality of IR'.

Source reference. Original lemma 02NS: sources/stacks-chow/tags.txt, source lines 3491–3491. Cross-chapter standard prerequisite invoked once at proof line 587 for projectivity of the blowup.

Source reference. Original lemma 0BFM: sources/stacks-chow/tags.txt, source lines 14390–14390. Cross-chapter standard prerequisite invoked once at proof line 589 for the location of the generic points of the blowup.

Source reference. Original lemma 0AB7: sources/stacks-chow/tags.txt, source lines 13010–13010. Cross-chapter standard prerequisite invoked once at proof line 590 for the finiteness criterion that makes the blowup affine.

Source reference. Original lemma 02M9: sources/stacks-chow/tags.txt, source lines 3436–3436. Cross-chapter standard prerequisite invoked once at proof lines 598-599 to upgrade locally principal to principal over the semi-local ring R'; the final writing a = a't, b = b't is retained in the counted span.

Lemma 0.1. *Let $(R, \mathfrak{m})$ be a Noetherian local ring of dimension 1. Let $a, b \in R$ be nonzerodivisors. There exists a finite ring extension $R \subset R'$ with R'/R annihilated by a power of $\mathfrak{m}$ and nonzerodivisors $t, a', b' \in R'$ such that $a = ta'$ and $b = tb'$ and $R' = a'R' + b'R'$.*

Proof. If a or b is a unit, then the lemma is true with $R = R'$. Thus we may assume $a, b \in \mathfrak{m}$. Set $I = (a, b)$. The idea is to blow up R in I. Instead of doing the algebraic argument we work geometrically. Let $X = \mathrm{Proj}(\bigoplus_{d \geq 0} I^d)$. By Divisors, Lemma 02OS the morphism $X \to \mathrm{Spec}(R)$ is an isomorphism over the punctured spectrum $U = \mathrm{Spec}(R) \setminus \{\mathfrak{m}\}$. Thus we may and do view U as an open subscheme of X. The morphism $X \to \mathrm{Spec}(R)$ is projective by Divisors, Lemma 02NS. Also, every generic point of X lies in U, for example by Divisors, Lemma 0BFM. It follows from Varieties, Lemma 0AB7 that $X \to \mathrm{Spec}(R)$ is finite. Thus $X = \mathrm{Spec}(R')$ is affine and $R \to R'$ is finite. We have $R_a \cong R'_a$ as $U = D(a)$. Hence a power of a annihilates the finite R-module R'/R. As $\mathfrak{m} = \sqrt{(a)}$ we see that R'/R is annihilated by a power of $\mathfrak{m}$. By Divisors, Lemma 02OS we see that IR' is a locally principal ideal. Since R' is semi-local we see that IR' is principal, see Algebra, Lemma 02M9, say $IR' = (t)$. Then we have $a = a't$ and $b = b't$ and everything is clear. $\square$